

Answer

$$dw = dx + dy$$

$$ds = 2x dx + 2y dy = 2dx + 2dy$$

$$dt = 2x dx - 2y dy = 2dx - 2dy$$

$$dx = \frac{1}{4} (ds + dt)$$

$$dy = \frac{1}{4} (ds - dt)$$

$$dw = \frac{1}{2} ds = \frac{\partial w}{\partial s} ds + \frac{\partial w}{\partial t} dt$$

$$\frac{\partial w}{\partial s} = \frac{1}{2} \quad \frac{\partial w}{\partial t} = 1$$

Answer

$$y'' + 5y' + 6y = 0 \quad \lambda_1 = -2 \quad \lambda_2 = -3$$

$$y_c(x) = c_1 e^{-2x} + c_2 e^{-3x}$$

Particular solution

1) for e^{-4x}

$$y_p' = c e^{-4x}$$

$$c \cdot (-4)^2 e^{-4x} + 5 \cdot c \cdot (-4) e^{-4x} + 6 \cdot c e^{-4x} = e^{-4x}$$

$$c \cdot [16 - 20 + 6] = 1$$

$$C(-4) + 70 = 1$$

$$C[16 - 20 + 6] = 1$$

$$C = \frac{1}{2}$$

$$y_p^1 = \frac{1}{2} e^{-4x}$$

2) for $\cos 5x$

consider $y_p^2 = A \cos 5x + B \sin 5x$

plug into the equation

$$A(-19) \cos 5x - 25A \sin 5x$$

$$+ B(-19) \sin 5x + 25B \cos 5x = \cos 5x$$

$$-19A + 25B = 1$$

$$25A + 19B = 0$$

$$\Rightarrow A = \frac{-19}{19^2 + 25^2} \quad B = \frac{25}{19^2 + 25^2}$$

3) for x ,
consider $y_p^3 = px + q$

$$p = \frac{1}{6} \quad q = -\frac{5}{36}$$

Final solution

$$y = y_c + y_p^1 + y_p^2 + y_p^3$$